

MAX SUN

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Education

University of Waterloo

Bachelor of Computer Engineering

2024 - Present

GPA: 3.77/4.0

Experience

Shopify

May 2026 - August 2026

Software Engineer Intern | Ruby, TypeScript, React Native

Toronto, Ontario

- Scaled Shopify's **mobile POS** for in-person merchants by optimizing **offline-to-online sync**, reducing 95th percentile sync latency by about 35% (1.7s to 1.1s) and supporting about **20k daily checkouts**.
- Built reliability and telemetry improvements for POS sessions in **React Native/TypeScript**, cutting crash rate by about 22% and increasing successful checkout sessions to about 99.3%.

BrainRidge Consulting

January 2026 – April 2026

Software Engineering Intern | TypeScript, Node.js, PostgreSQL, AWS, GitHub Actions

Toronto, Ontario

- Designed and deployed a highly-scalable automated HTTP data ingestion pipeline (**Node.js**) for a **RAG-based AI model**, accelerating knowledge base expansion by 70% and **eliminating 83% of manual administrative overhead**.
- Spearheaded the development of a multi-query **RAG-Fusion retrieval engine** utilizing **AWS Bedrock** and advanced LLM prompt engineering, achieving an unprecedented **98% document recall** for ambiguous user queries.
- Architected a **hybrid search infrastructure** combining **BM25** and **vector search** with **Reciprocal Rank Fusion** within **PostgreSQL**, optimizing exact-match retrieval accuracy and plummeting zero-result rates by 62%.
- Pioneered a real-time, low-latency **conversational AI voice system** via **WebSockets** and **AWS Nova**, delivering sub-500ms median response times for natural banking interactions complete with voice barge-in capabilities.
- Transformed cloud infrastructure deployment by engineering a multi-environment **CI/CD pipeline** leveraging **Terraform, GitHub Actions, and AWS**, accelerating release velocity by 73% with rigorous production safety gates.

Exo-Insights

May 2025 – August 2025

Software Developer Intern | Android Studio, Java/Kotlin

Waterloo, Ontario

- Engineered a robust **Android SDK (Java/Kotlin)** to ingest and process high-frequency (25Hz) gaze-tracking telemetry from smart glasses, with **99.9% data upload reliability** via RESTful APIs across 30+ edge devices.
- Developed high-performance **Android UI components** for real-time biometric visualization (heart-rate and gaze tracking), optimizing internal QA testing lifecycles and **reducing debugging overhead by 30%**.
- Streamlined sensor-to-cloud data ingestion by integrating custom **Bluetooth SDKs** and **REST APIs** into the core pipeline, significantly accelerating data availability by **cutting latency by 40%**.

Projects

Tower of Wizard | Unity, C#

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- Architected and commercially launched 'Tower of Wizard,' a high-performance 2D action roguelike game (Unity, C#), driving a successful go-to-market strategy that secured **1,000+ wishlists** and **150,000+ impressions**.
- Re-engineered core gameplay physics and logic from standard OOP to **Unity DOTS/ECS** (Entity Component System), unlocking a **500% increase in maximum on-screen entities (10k+)** while slashing frame times by 35%.
- Integrated the **Steamworks SDK** to engineer cross-platform cloud saves, secure achievement tracking, and workshop distribution systems, tangibly **lifting user retention by 12%** within the first month.

Hack the North 2025: EyeCandy | TypeScript, Lens Studio, Gemini

[View GitHub](#)

- Spearheaded the development of an award-winning AR shopping platform (TypeScript, Lens Studio) integrating live **Shopify API** data, securing the **'Runner-Up Best Spectacles Project'** out of 1,000+ hackers.
- Engineered the spatial interaction layer utilizing full-body tracking, 3D model rendering, and **Google Gemini LLM** voice search, delivering seamless, hands-free product try-ons with sub-2-second end-to-end latency.

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C++, C, C#, Ruby, SQL, GraphQL

Full Stack & Backend: React, React Native, Next.js, Node.js, Express, PostgreSQL, MongoDB, Redis, WebSockets

Cloud & Infrastructure: AWS (Bedrock, S3, Nova), Docker, Terraform, Kubernetes, GitHub Actions, CI/CD, Git

AI & Specialized: RAG (Retrieval-Augmented Gen), LLM Prompt Engineering, Vector Search, Android SDK, ARKit